transfers carbohydrates to the fungus, while the fungus transfers limiting nutrients, such as phosphorus, to the plant. Each partner benefits, but each partner also experiences a cost: It transfers materials that it could have used to support its own growth and metabolism. For an interaction to be considered a mutualism, the benefits to each partner must exceed the costs. When this is not the case, the mutualism may break down, at least temporarily. In some mycorrhizae, for example, the plant may cease to supply carbohydrates to its fungal partner when soil nutrients are plentiful—a change that occurs because the cost of supporting the fungus has become greater than the benefits the fungus can provide.

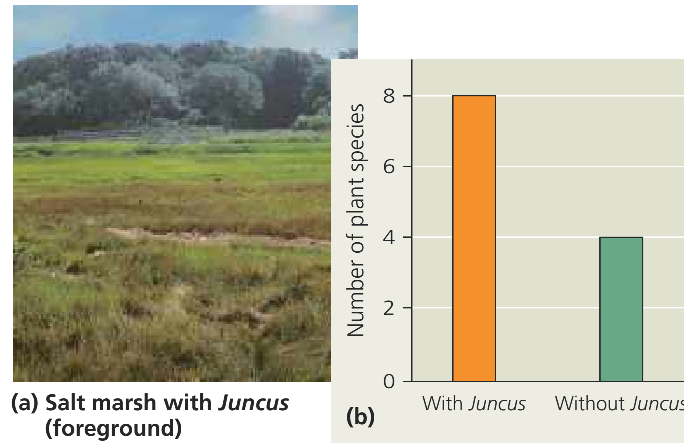
Commensalism

An interaction that benefits the individuals of one of the interacting species but neither harms nor helps the individuals of the other species is called **commensalism** (+/0). Like mutualism, commensalisms are common in nature. For instance, many wildflowers that grow best in low light levels are found only in shaded, forest floor environments. Such shade-tolerant "specialists" depend entirely on the trees that tower above them—the trees provide their dim habitat. Yet the survival and reproduction of the trees are not affected by these wildflowers. Thus, these species are involved in a +/0 interaction in which the wildflowers benefit and the trees are not affected.

In another example of a commensalism, cattle egrets feed on insects flushed out of the grass by grazing bison, cattle, horses, and other herbivores **(Figure 54.10)**. Because the birds increase their feeding rates when following the herbivores, they clearly benefit from the association. Much of the time, the herbivores are not affected by the birds. At times, however, the herbivores too may derive some benefit; for example, the birds may remove and eat ticks and other ectoparasites from the herbivores' skin, or they may warn the herbivores of a predator's approach. This example provides another illustration of a key point about ecological interactions: Their effects can change over time. In this case, an interaction whose effects are typically +/0 (commensalism) may at times become +/+ (mutualism).

▼ **Figure 54.10 Commensalism between cattle egrets and African buffalo.**

▼ **Figure 54.11 Positive interactions in New England salt marshes.** When black rush (*Juncus gerardii*) is present, soil salt concentrations drop and soil oxygen levels rise, increasing the number of plant species that can live in the marsh.



(a) Salt marsh with *Juncus* (foreground)

(b)

Positive interactions can have major effects on ecological communities. For instance, the black rush *Juncus gerardii* alters soil conditions in ways that benefit other plant species in New England salt marshes **(Figure 54.11a)**. *Juncus* shades the soil surface, which reduces evaporation and thus lowers the concentration of salt in the soil. The presence of *Juncus* also increases soil oxygen levels; this occurs because some oxygen leaks into the soil as *Juncus* transports oxygen to its belowground tissues. In one study, when *Juncus* was removed from areas of the marsh, those areas supported 50% fewer plant species **(Figure 54.11b)**.

Like positive interactions, competition and exploitation (predation, herbivory, and parasitism) also can strongly affect ecological communities, as examples throughout the rest of this chapter will show.

CONCEPT CHECK 54.1

1. Explain how competition, predation, and mutualism differ in their effects on members of the two interacting species.
2. According to the principle of competitive exclusion, what outcome is expected when two species with identical niches compete for a resource? Why?
3. **MAKE CONNECTIONS** Figure 24.14 illustrates how a hybrid zone can change over time. Imagine that two finch species colonize a new island and are capable of hybridizing (mating and producing viable offspring). The island contains two plant species, one with large seeds and one with small seeds, growing in isolated habitats. If the two finch species specialize in eating different plant species, would reproductive barriers be reinforced, weakened, or unchanged in this hybrid zone? Explain.

For suggested answers, see Appendix A.